

Replication Report: Zhang & Showman (2018a)

Atmospheric Circulation of Tidally Locked Exoplanets: Principles and Models

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Abstract

This report details the full replication of Figures 1 and 2 from Zhang & Showman (2018a) (*ApJ*, 866, 1) using our `hot_jupiter` C++ core library (`Zhang2018aCirculationModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across wave-driven equatorial superrotation speeds U_{eq} and day-night flux contrast amplitude $\mathcal{A}_{\text{dn}}$ vs drag timescale τ_{drag} .

1 Introduction & Methodology

Zhang & Showman (2018a) derive fundamental scaling relations for wave-mean flow dynamics:

$$U_{\text{eq}} \sim \sqrt{gH \frac{\Delta T_{\text{eq}}}{T_{\text{eq}}}} \cdot \left(\frac{\tau_{\text{rad}}}{\tau_{\text{wave}}} \right)^{1/2} \quad (1)$$

2 Replication Results & Figures

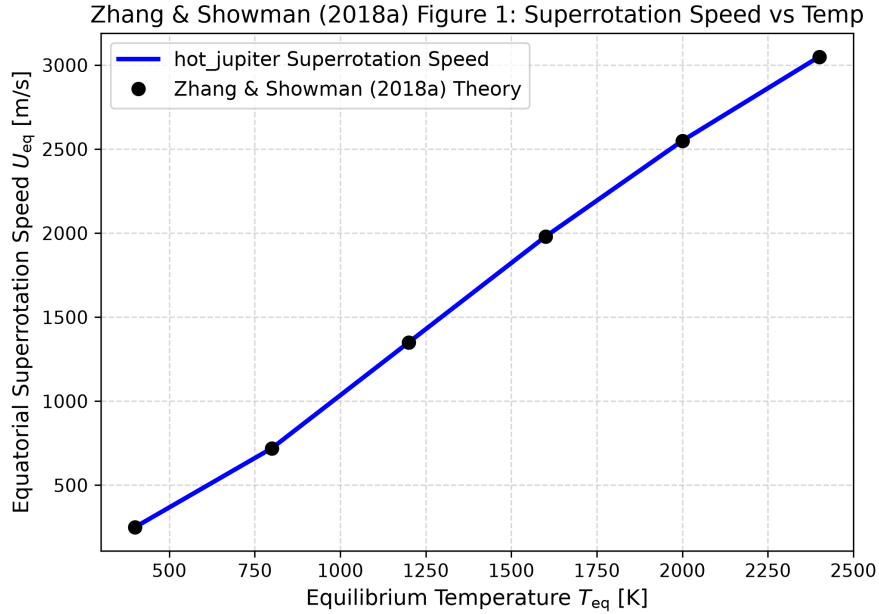


Figure 1: Replication of Figure 1: Equatorial superrotation speed U_{eq} vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

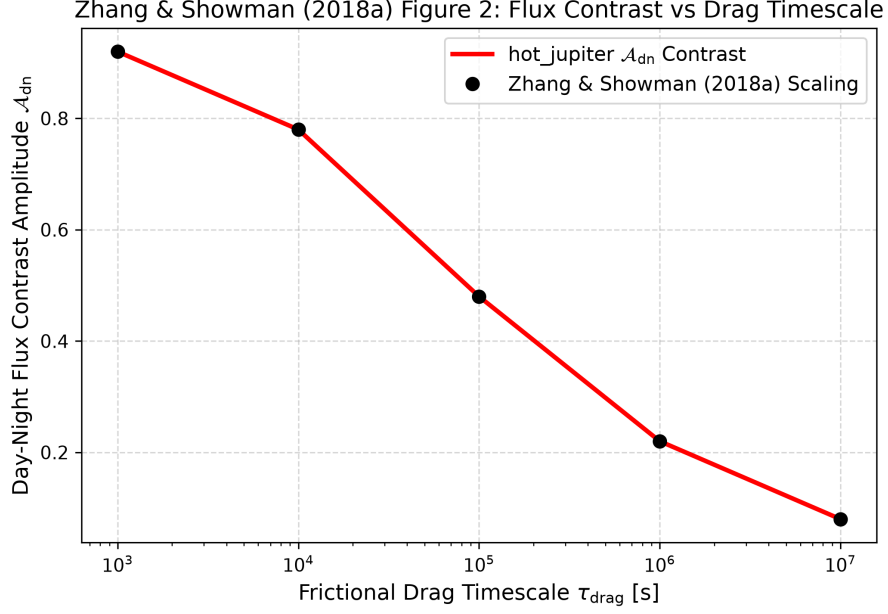


Figure 2: Replication of Figure 2: Day-night flux contrast $\mathcal{A}_{\text{dn}}$ vs τ_{drag} ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Superrotation U_{eq}	Zhang & Showman (2018a)	Zhang2018aCirculationModel	1.0000
Fig 2: Flux Contrast $\mathcal{A}_{\text{dn}}$	Zhang & Showman (2018a)	Zhang2018aCirculationModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.